Topic 2: Network Models - OSI & TCP/IP

Course outline weeks 1-2 (OSI reference model, services) + weeks 5-9 (layers). The single MOST examined topic: in all 5 papers, often 9-15 marks.

How often this topic is examined

Question asked	Paper(s)	Marks
List the 7 OSI layers in order (from layer 7)	2025 Q1a.i	9
Purpose of the OSI model + functions of Layer 3 and Layer 4	2025 Q1a.ii	8
Labelled OSI diagram + function of each layer	SE2024 Q3b [9], ISA2024 Q3c [15]	9-15
OSI vs TCP/IP (compare & contrast)	2018 Q5b [12], 2023 Q1c [4]	4-12
Significance of TCP/IP and its implementation	SE2024 Q1e	4
IP vs MAC address - which OSI layer each belongs to	2025 Q5c	6
Router vs switch - OSI layers and functionality	2025 Q5d	10 (see Topic 4)
Internetworking principles; how TCP/IP and IP packets help	2025 Q5a	3

EXAM TIP: Learn the OSI table on page 2 by heart: layer number, name, PDU, function, protocols, devices. With that one table you can answer 9, 12 or 15-mark questions.

1. The OSI reference model

The **Open Systems Interconnection (OSI)** model was developed by **ISO** (1984). It is a 7-layer **conceptual framework** that describes how data moves from an application on one computer, across a network, to an application on another computer.

Purpose of the OSI model (2025 Q1a.ii)

- Provides a **standard** so that equipment and software from different vendors can **interoperate**.
- **Divides the complex** communication process into smaller, manageable layers - each layer has a defined job and only talks to the layer above and below (modularity).
- A change in one layer does not affect the others (e.g. move from copper to fibre without changing the applications).
- Helps **troubleshooting** (work layer by layer: 'is it a Layer 1 cable problem or a Layer 3 IP problem?') and **teaching**.

REMEMBER: Mnemonic top-down (7->1): **All People Seem To Need Data Processing** = Application, Presentation, Session, Transport, Network, Data Link, Physical. Bottom-up (1->7): **Please Do Not Throw Sausage Pizza Away**.

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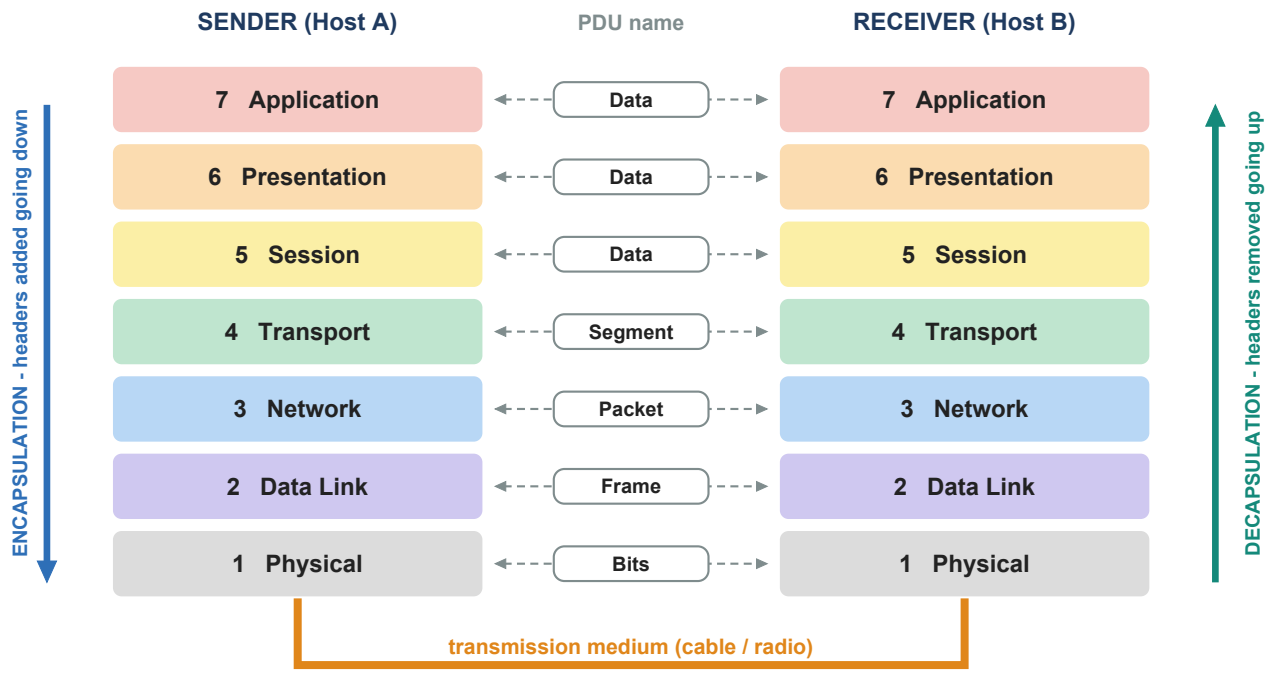
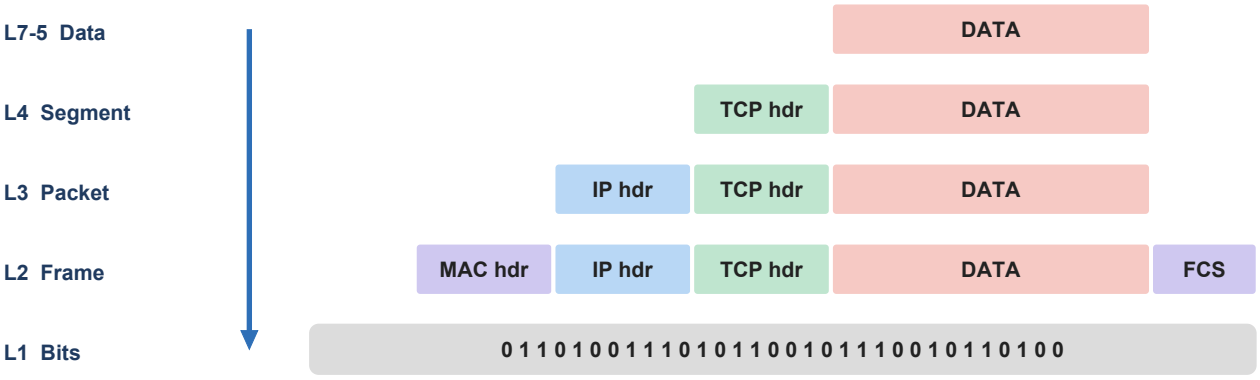


Figure: the OSI model - peer layers, PDUs and encapsulation



each layer wraps the one above: header added in front (Data Link also adds a trailer = FCS/CRC)

Figure: encapsulation - each layer adds its header

The OSI table - learn this

Layer	PDU	Main functions	Protocols / examples	Devices
7 Application	Data	Interface between user/applications and the network; network services: file transfer, email, web, remote login, directory services	HTTP, HTTPS, FTP, SMTP, POP3, IMAP, DNS, DHCP, Telnet, SSH, SNMP	Gateway, hosts
6 Presentation	Data	Translation (data formats, ASCII/EBCDIC), **encryption/decryption**, **compression**; 'syntax and semantics' of data	SSL/TLS, JPEG, GIF, MPEG, ASCII	-
5 Session	Data	Establish, maintain & terminate sessions (dialog control - half/full duplex); **synchronisation/checkpoints** so a long transfer can resume; RPC	NetBIOS, RPC, PPTP, SQL sessions	-

Layer	PDU	Main functions	Protocols / examples	Devices
4 Transport	Segment (TCP) / Datagram (UDP)	**Process-to-process (end-to-end)** delivery using **port numbers**; segmentation & reassembly; connection control; **flow control**; **error control** (end-to-end) & retransmission	TCP, UDP, SCTP	Firewalls (L4)
3 Network	Packet	**Logical addressing (IP)**; **routing** - choosing the best path across networks (source-to-destination, host-to-host delivery); fragmentation; congestion control	IPv4, IPv6, ICMP, ARP*, OSPF, RIP, EIGRP	**Router**, L3 switch
2 Data Link	Frame	**Framing**; **physical (MAC) addressing**; **hop-to-hop (node-to-node)** delivery; error detection (CRC); flow control; **media access control** (CSMA/CD). Sub-layers LLC + MAC	Ethernet, Wi-Fi (802.11), PPP, HDLC	**Switch**, bridge, NIC
1 Physical	Bits	Transmits raw **bits** over the medium: physical characteristics of interfaces & media, signal/encoding, data rate, bit synchronisation, line configuration, topology, transmission mode	RJ45, Ethernet cabling, DSL, USB, Bluetooth radio	**Hub**, repeater, cables, modem

*ARP sits between layers 2 and 3 (it maps a Layer-3 IP address to a Layer-2 MAC address).

2. Model answer: Functions of Layer 3 and Layer 4 (2025 Q1a.ii)

Layer 3 - Network layer

- **Logical addressing** - adds source and destination IP addresses to every packet so hosts on different networks can be identified.
- **Routing** - routers use routing tables/protocols (RIP, OSPF) to select the best path from source network to destination network.
- **Source-to-destination delivery** of packets across multiple networks (internetworking).
- **Fragmentation & reassembly** of packets to fit the MTU of each link.
- **Congestion control** and error reporting (ICMP).

Layer 4 - Transport layer

- **Process-to-process delivery** - uses **port numbers** (e.g. HTTP 80, HTTPS 443) to deliver data to the correct application.
- **Segmentation and reassembly** - splits messages into segments with sequence numbers, reassembles them in order at the receiver.
- **Connection control** - connection-oriented (TCP 3-way handshake) or connectionless (UDP).
- **Flow control** - stops a fast sender from overwhelming a slow receiver (sliding window).
- **Error control** - end-to-end; detects lost/damaged segments using checksums & ACKs and retransmits them.

3. The TCP/IP model (Internet model)

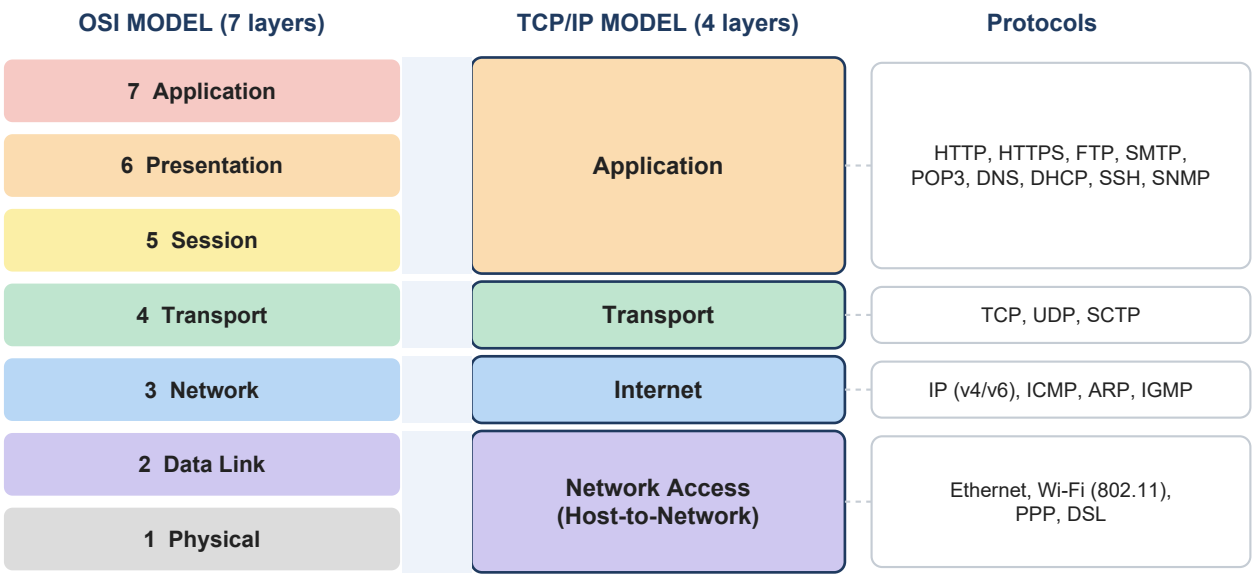


Figure: OSI layers mapped to the TCP/IP model

Significance of TCP/IP and its implementation (SE2024 Q1e)

- It is **the protocol suite of the Internet** - any device that runs TCP/IP can talk to any other on the Internet regardless of hardware or OS (**interoperability**, open standard, vendor-neutral).
- **Routable & scalable** - IP addressing and routing let millions of networks interconnect.
- **Reliable delivery** via TCP (ACKs, retransmission) and fast delivery via UDP for real-time traffic.
- **Implementation:** built into every OS's network stack; each host is configured with an **IP address, subnet mask, default gateway and DNS server** (manually or via DHCP); applications use **sockets** (IP + port) to communicate.

4. OSI vs TCP/IP - compare and contrast

EXAM QUESTION 2018 Q5b [12], 2023 Q1c [4] **[4-12]**
Differentiate between the OSI model and the TCP/IP protocol suite. / Compare and contrast the key differences between the OSI Layer model and the TCP/IP Reference model.

Point	OSI model	TCP/IP model
Developed by	ISO (International Organization for Standardization)	US Department of Defense (DARPA) / IETF
Number of layers	7 layers	4 layers (or 5)
Nature	Theoretical / **reference model** - a generic, protocol-independent standard	**Practical**, implemented model - the protocols came first and the model describes them
Approach	Model defined **before** protocols (top-down)	Protocols defined first, model fitted later
Separate presentation & session layers	Yes	No - merged into the Application layer
Physical & data link	Two separate layers	Combined into Network Access layer
Transport layer	Connection-oriented service guaranteed	Both connection-oriented (TCP) and connectionless (UDP)

Point	OSI model	TCP/IP model
Network layer	Supports connection-oriented and connectionless	Connectionless only (IP)
Distinction of service, interface, protocol	Clearly defined	Not clearly separated
Usage	Used for teaching and troubleshooting	Used on the real Internet
Flexibility	Protocols are better hidden, easy to replace	Hard to replace protocols

Similarities (for 'compare'): both are layered; both have application, transport and network layers with similar functions; both use packet switching; both use encapsulation; transport layer provides end-to-end delivery in both.

EXAM TIP: For 12 marks give ~6 differences in a table (2 marks each: one for OSI side, one for TCP/IP side) + draw the two models side by side.

5. IP address vs MAC address

EXAM QUESTION 2025 Q5c **[6]**
Explain the difference between IP addresses and MAC addresses. Which OSI layers does each address belong to?

Feature	IP address	MAC address
OSI layer	**Layer 3 - Network**	**Layer 2 - Data Link (MAC sub-layer)**
Type	**Logical** address	**Physical / hardware** address
Size	IPv4 32 bits (e.g. 192.168.1.10); IPv6 128 bits	48 bits, 12 hex digits (e.g. 00:1A:2B:3C:4D:5E)
Assigned by	Network admin / DHCP / ISP; can change when you move networks	Burned into the NIC by the manufacturer (OUI from IEEE); normally permanent
Scope / purpose	Identifies a host **and its network**; used for **routing** between networks (end-to-end)	Identifies a NIC on the **local** network; used for **hop-to-hop** delivery in a LAN
Changes during the journey?	Source & destination IP stay the same end-to-end	Changes at every hop (each router rewrites the frame)
Resolved by	DNS maps names -> IP	ARP maps IP -> MAC

Third type for completeness: **port address** (Layer 4, 16-bit) identifies the process/application, e.g. 80 for HTTP.

6. Internetworking principles (2025 Q5a, outline weeks 5-7)

Internetworking = connecting separate, possibly different networks (Ethernet LANs, Wi-Fi, WAN links) so they act as one large network (the Internet = network of networks).

- **Universal addressing** - every host gets a unique IP address independent of the underlying LAN technology.
- **Routers** connect networks and forward packets hop-by-hop using routing tables.

- **Common protocol (IP)** - every network carries the same IP packet format, hiding hardware differences; **TCP** above it provides reliable end-to-end delivery.
- **Encapsulation** - the IP packet is placed inside whatever frame each network uses (Ethernet, PPP); **fragmentation** adapts packet size to each network's MTU.
- **Connectionless, best-effort datagram delivery** - each IP packet is routed independently.

REMEMBER: Delivery types: **Data Link = hop-to-hop (node-to-node)**, **Network = host-to-host (source-to-destination)**, **Transport = process-to-process (end-to-end)**. Examiners love this distinction.

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